Ex:06.
3.9.26

HAMMING CODE

AIM:
To write code for finding Hamming code in python.

CODE:

```
data = list(map(int, input("Enter data bits")))
m = len(data)
for i in range(0,4):
r=0 if (2**i
while 2**r < m+r+1:
    r+=1
n=m+r
code = [0]*(n+1)  # code => list n+1 ele all 0 initializing
j=0
for i in range(1,n+1):
    if i & (i-1)==0:
        code[i]=0
    else:
        code[i]= data[j]
        j+=1

# calculate parity bits  (make this a fn.)
  for i in range(r):
      pos = 2**i
      par =0
      for j in range(1,n+1):
          if j & pos and j != pos:
              par ^= code[j]
      code[pos] = par

print("Hamming code:") end=" "
 for i in range(n, 0, -1):
     print(code[i], end=" ")
```

```
# Error
error = int(input("\n Enter error position (0=no error)"))
if error != 0:
    code[error] ^= 1
print("Received code:")
for i in range(n, 0, -1):
    print(code[i], end=" ")
errorpos = 0
for i in range(r):
    pos = 2**i
    par = 0
    for j in range(1, n+1):
        if j & pos:
            par ^= code[j]
    if par:
        errorpos += pos
if errorpos == 0:
    print("NO error")
else:
    print("Error at:", errorpos)
    code[errorpos] ^= 1
    print("corrected code:")
    for i in range(n,0,-1):
        print(code[i], end=" ")
print()
```

RESULT: Thus, the code for finding hamming code was executed successfully.